



Society of Petroleum Engineers

SPE-229914-MS

Application of Sweep Control and Targeted Oil Displacement Technology in Low-Permeability Reservoirs - A Case Study of a Reservoir in Western China

G. Shi, Startwell Beijing Technology Co. Ltd, Beijing, China; X. Wu, Startwell Energy Co Ltd, Beijing, China; J. Wang, Y. Shang Guan, J. Wu, W. Zheng, K. Cheng, and X. Zhu, CNPC Changqing Oil Field Co, Xian, China

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229914-MS](https://doi.org/10.2118/229914-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The Y19 Oilfield, a low-permeability reservoir in western China, is currently in the high water-cut stage of waterflooding and suffers from severe non-uniform waterflooding. To address this issue, a novel enhanced oil recovery (EOR) technology has been applied to optimize injected water flow, improve sweep and displacement efficiency, and boost the oil recovery factor.

In addressing the severe uneven waterflooding in this high-water-cut low-permeability reservoir, a sweep control & targeted oil displacement technology integrating a three-component composite system (highly viscous active polymer (HVAP), soft microgel (SMG), and nano-intelligent emulsion (NanolE)) is applied. The methodology involves a sequential and synergistic workflow: 1. HVAP plugs high-permeability zones by leveraging its delayed viscosity-enhancing properties, reducing excessive water channeling; 2. SMG injection follows, utilizing its elastic deformability to achieve deep-profile modification and redirecting injected fluids into low-permeability regions; 3. NanolE incorporation reduces interfacial tension and alters rock wettability, enhancing residual oil mobilization in target zones. This integrated approach creates a "blocking-deep diversion-efficiency enhancement" mechanism, which systematically resolves two key challenges: macroscopic sweep imbalance and microscopic displacement inefficiency.

This paper presents a novel sweep control & targeted oil displacement technology to address the uneven waterflooding in high-water-cut low-permeability reservoirs. Laboratory experiments show that the oil displacement efficiency of natural cores reaches 28.6% and that of artificial cores reaches 29.1%. Numerical simulations indicate an enhanced oil recovery (EOR) of 10.1%, with HVAP contributing 3.28%, SMG 2.8%, and NanolE 4.02%. Economic evaluation reveals a favorable input – output ratio of 1:3.6 at an oil price of \$60/bbl. Field tests initiated in August 2024 showed significant results within ten months. The average daily oil production of central affected wells increased by 54.2%, and the water cut decreased by 14.1%. This technology synergistically improves sweep and displacement efficiency by plugging dominant channels, reconstructing the seepage field, and optimizing the mobility ratio. Compared with traditional polymer flooding, it offers better reservoir adaptability and economic viability, providing an innovative solution for high-water-cut development in low-permeability sandstone reservoirs.

Keywords: Sweep control, targeted oil displacement, seepage field regulation, low-permeability reservoir, enhanced oil recovery

Introduction

Basic Reservoir Overview

Y19 Oilfield is a typical fluvial facies sedimentary lithological-structural reservoir, with the following reservoir parameters: temperature of 45°C, total salinity of 11500 mg/L, crude oil viscosity of 5.32 mPa·s; average porosity of 17.2%, average permeability of 15.1 mD, belonging to a low-porosity and low-permeability reservoir.

The oilfield was put into development in 2008, adopting a triangular well pattern waterflooding development. At present, it has entered the "dual-high stage" of "high water cut and high recovery degree", with prominent waterflooding contradictions: low water sweep efficiency, rising water consumption rate coupled with declining water retention rate, and intensified ineffective circulation of injected water, leading to accelerated increase in water cut and deteriorated development performance; after water breakthrough in oil layers, the remaining oil is scattered in distribution, making it increasingly difficult to tap potential through conventional primary well pattern waterflooding.

Historical conformance control measures show that existing technologies are difficult to break through the bottleneck: microsphere deep conformance control implemented in 20 wells from 2016 to 2020 failed to curb production decline and water cut increase; viscoelastic profile control carried out in 14 well groups in 2021 resulted in a short-term increase of 8 tons in daily oil production, but the effective period was only 3 months; large-scale profile control combined with local infill adjustment from 2019 to 2022 temporarily slowed down the rate of water cut increase, but continuous injection after optimizing system parameters from 2022 to 2023 still failed to achieve the expected effect. In addition, since 2015, development indicators have further deteriorated due to factors such as excessive liquid production intensity and casing damage.

In summary, the Y19 oilfield urgently requires new, efficient improved oil recovery/enhanced oil recovery (IOR/EOR) technologies to overcome its development challenges. Against this backdrop, the "Sweep Control and Targeted Oil Displacement Technology" has been proposed as a tailored solution, with its technical concepts, principles, and applicability outlined in Section 1.2 below.

Sweep Control and Targeted Oil Displacement Technology

Technical Concept and Connotation. Sweep control and targeted oil displacement technology centers on the overall regulation of the entire flow field system. By optimizing the combination mode and injection sequence of chemical agents, and integrating various new functional materials, it achieves effective sweep at the micro-pore scale, thereby realizing efficient production of remaining oil and efficient displacement of residual oil.

The core of this technology lies in the systematic integration of various sweep technologies — based on expanding sweep volume, it accurately covers the effective sweep range, and ultimately significantly improves oil displacement efficiency while ensuring sweep efficiency. Compared with traditional technologies, it features prominent characteristics of proactive regulation, aggressive displacement, targeted action, and high-efficiency production, forming a full-scale synergistic mechanism from macro flow field regulation to micro displacement enhancement.

Technical Principle. Viscous fluids (e.g., polymers or cross-linked polymer gels) adjust the mobility ratio by increasing the viscosity of the displacement medium^[1-3]. These fluids tend to flow preferentially into high-permeability large pores: while they can raise aqueous phase viscosity and reduce its relative permeability, their high viscosity significantly increases seepage resistance in low-permeability layers. In some cases, they may even fail to penetrate these layers, leaving remaining oil in low-permeability zones unswept^[4-8]. In

this case, the oil phase cannot be effectively mobilized, and its relative permeability remains at a low level. Analyzed from the classic mobility ratio (M) formula, viscous fluids struggle to achieve balanced sweep of both high and low permeability layers, and the effect of mobility ratio optimization is limited, ultimately resulting in unsatisfactory displacement efficiency.

$$M = \frac{\mu_o}{k_o} \cdot \frac{k_w}{\mu_w}$$

Equation 1—Equation of Mobility Ratio.

The dispersed-phase displacement medium Soft Microgel (SMG) has a viscosity slightly higher than that of water, and its mobility ratio regulation does not rely on increasing its own viscosity. SMG can directly reduce the relative permeability of the aqueous phase in large pores by preferentially temporarily plugging the pore throats of preferential water flow channels in deep formation^[9-10-11]. Meanwhile, due to its low viscosity, it does not significantly increase the injection resistance in low-permeability layers, allowing subsequent driving fluids (e.g., injected water) to smoothly enter low-permeability zones. In synergy with nano oil displacement agents, it achieves efficient production of remaining oil and effective mobilization of residual oil, significantly increasing the relative permeability of the oil phase and thus realizing efficient improvement of the mobility ratio^[12].

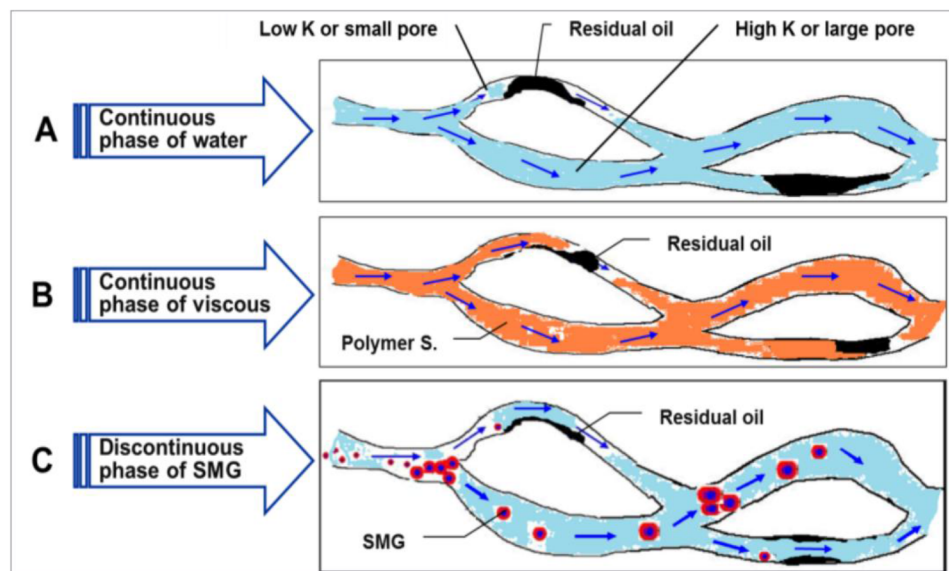


Figure 1—Comparison of displacement mechanisms between continuous-phase fluids and dispersed-phase fluids (A: schematic diagram of low efficiency in water flooding; B: plugging limitations of traditional polymer flooding; C: advantages of SMG dispersed-phase displacement).

The Sweep Control and Targeted Oil Displacement Technology integrates the technical advantages of continuous-phase displacement and dispersed-phase displacement to establish a displacement system featuring "segmented regulation and synergistic enhancement":

1. Utilizing the mobility control capability of continuous-phase viscous fluids (e.g., HVAP), it prioritizes plugging or suppressing channeling in preferential water flow channels, achieving preliminary confinement of the macroscopic sweep range and laying a foundation for subsequent deep displacement.
2. Dispersed-phase fluids (e.g., SMG) are injected, leveraging their characteristics of low viscosity and strong penetrability to directionally enter deep low-permeability zones or remaining oil-rich areas in

the reservoir, targetedly displacing residual oil that is difficult to be swept by conventional continuous phases.

3. Through the microscopic enhancement effect of nano oil displacement agents (e.g., NanoIE) (such as reducing interfacial tension), the oil-water displacement efficiency is further improved, realizing full-scale optimization from the expansion of macroscopic sweep volume to the enhancement of microscopic oil displacement efficiency, thus maximizing oil recovery.

By virtue of temporal coordination and functional complementarity of multiphase media, this technology resolves the contradiction between "macroscopic channeling" and "microscopic inefficiency" in single-phase displacement, forming an integrated oil displacement mechanism covering flow field regulation, deep displacement, and interface enhancement.

Technical Applicability Scope. Table 1 presents the reservoir applicability conditions for the Sweep Control and Targeted Oil Displacement Technology. Compared with single EOR technologies such as polymer flooding, surfactant flooding, and microbial flooding, this technology has broader universality. Its core advantage lies in that, through dynamically optimizing chemical agent combinations and slug designs, it can flexibly adapt to waterflood reservoirs with different geological conditions and development stages, thus achieving effective coverage of most waterflood reservoirs^[13]. This characteristic of multi-technical module integration breaks through the strict restrictions of single displacement technologies on parameters such as reservoir temperature, salinity, and crude oil properties, providing a more adaptable solution for enhancing oil recovery in complex reservoirs.

Table 1—Reservoir Adaptability of the Sweep Control and Targeted Oil Displacement Technology.

NO.	Item		Parameter	Optimal Parameter
1	Static data	Reservoir rock type	Clastic rock	Clastic rock
2		Reservoir depth, m	<4000	<3000
3		Reservoir thickness, m	>1	>1
4		Formation water	Unbounded bottom water or poorly developed bottom water	Boundless bottom water
5		Effective porosity, %	10~30	15~30
6		Permeability, mD	1~3000	10~500
7		pH value	7~9	7~8
8		Formation temperature, °C	30~130	40~100
9		Reservoir pressure, MPa	<50	≤25
10		Crude oil viscosity, cP	<200	<50
11		Salinity of formation water, mg/L	<300000	5000~100000
12		Divalent cations, mg/L	<3000	<500
13	Dynamic data	Remaining oil saturation, %	>30	30~50
14		Recovery factor, %	≤60	≤30
15		Water cut, %	>20	40~90
16		Injection-production well space, m	<700	100~300
17	Site injection	Fluid preparation water	Fresh water or sewage	Fresh water
18		Injection equipment	skid-mounted or pump station	skid-mounted or pump station

Main chemical agents. Based on the geological and development characteristics of Y19 Oilfield, high viscous active polymer (HVAP), soft microgel (SMG), and nano-intelligent emulsion (NanoIE) are selected as the core chemical agent system for the test.

- HVAP is a novel type of soluble polymeric material prepared by introducing temperature-resistant functional groups, salt-resistant functional groups, and active functional groups into polyacrylamide as the base. It possesses both the properties of polymers and surfactants.
- SMG is a polymeric emulsion with polyacrylamide-based polymers and surfactants as the main components, featuring slow-swelling properties. It has a particle size range of 30 nm to 600 μm , and the particle size can be customized according to the target reservoir.
- NanoIE exhibits strong capabilities in crude oil emulsification, wettability alteration, and in-situ scale self-growth regulation of heterogeneity. In particular, the viscosity of the injected fluid increases within a certain range as the water saturation in reservoir pores rises, forming a viscosity-increasing fluid to achieve self-controlled mobility and synergistically enhance microscopic oil displacement.

Table 2 presents the characteristics, and applicable scopes of each chemical agent.

Table 2—Characteristics and Application Objectives of Different Chemical Agents.

Chemical agent system	Distinctive features	Target audience
High Viscosity Active Polymer (HVAP)	HVAP exhibits higher viscosity (10 to 20 times) than traditional HPAM, with properties of delayed thickening and emulsification.	Blocking advantageous flow channels and suppressing seepage channels
Soft microgel (SMG)	Adjustable particle size, suitable for different pore throat formations, improving sweep efficiency	Pore-level flow channels, residual oil accumulation zones
Nano Intelligent Emulsion (NanoIE)	Lower relative permeability/good oil displacement performance	Residual oil enrichment zone, targeted oil displacement

Laboratory Experiments

HVAP Performance Evaluation Experiments

Plugging Performance Experiment. HVAP is used to plug preferential channels. Cores with gas-measured permeability of 200 to 300 mD were selected to represent preferential water flow channels. Table 3 shows the test results of sand-packed tube permeability, Table 4 presents the test results of sand-packed tube plugging rate, and Fig. 2 shows the HVAP injection curves of 1# and 2# sand-packed tubes. The experimental results indicate that the plugging rate of HVAP for cores exceeds 97%.

Table 3—Test Data of Sand-Packed Tube Permeability.

Sand filling pipe Number	Injection rate, ml/min	Inlet pressure, KPa	Outlet pressure, KPa	Permeability, mD	Pore volume, cm3
1#	1	45.1	9.4	204	71.2
2#	1	23.4	6.8	338	78.0

Table 4—Test Data of Sand-Packed Tube Plugging Rate.

Sand filling pipe Number	Injection rate, ml/min	Permeability, mD	Maximum breakthrough pressure corresponding to permeability, mD	Maximum breakthrough pressure corresponding to blockage rate, %
1#	1	204	3.4	98.4
2#	1	338	7.9	97.7

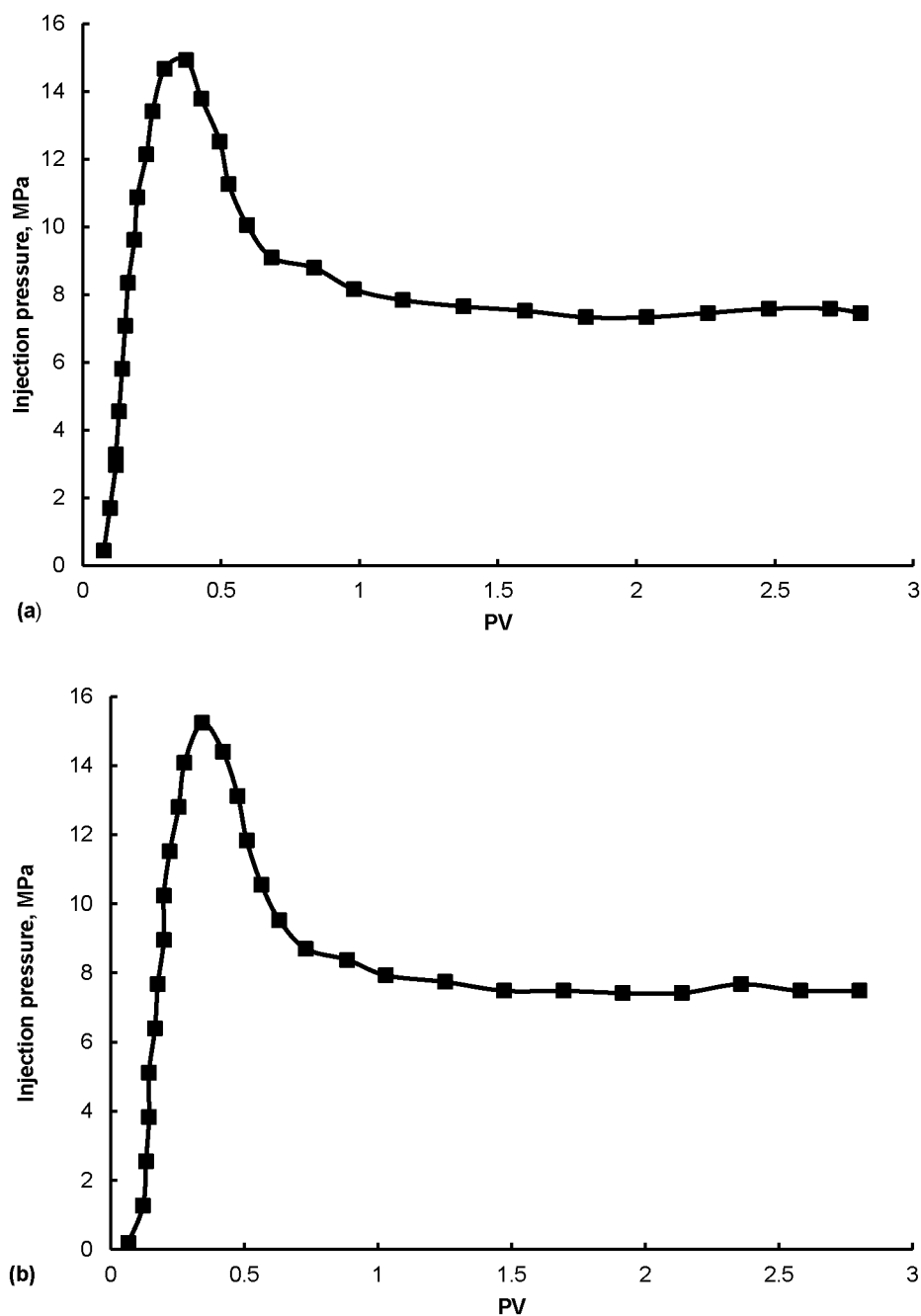


Figure 2—HVAP injection curves of sand-packed tubes (a) 1# sand-packed tube; (b) 2# sand-packed tube.

Viscosity-increasing and shear resistance experiments. HVAP solution with a concentration of 2000 mg/L was prepared using produced water from Y19 Oilfield, and its viscosity change over time was tested (Fig. 3). The results confirm that HVAP has excellent viscosity-increasing performance and maintains high viscosity over time—validating its role as a key component for plugging high-permeability zones.

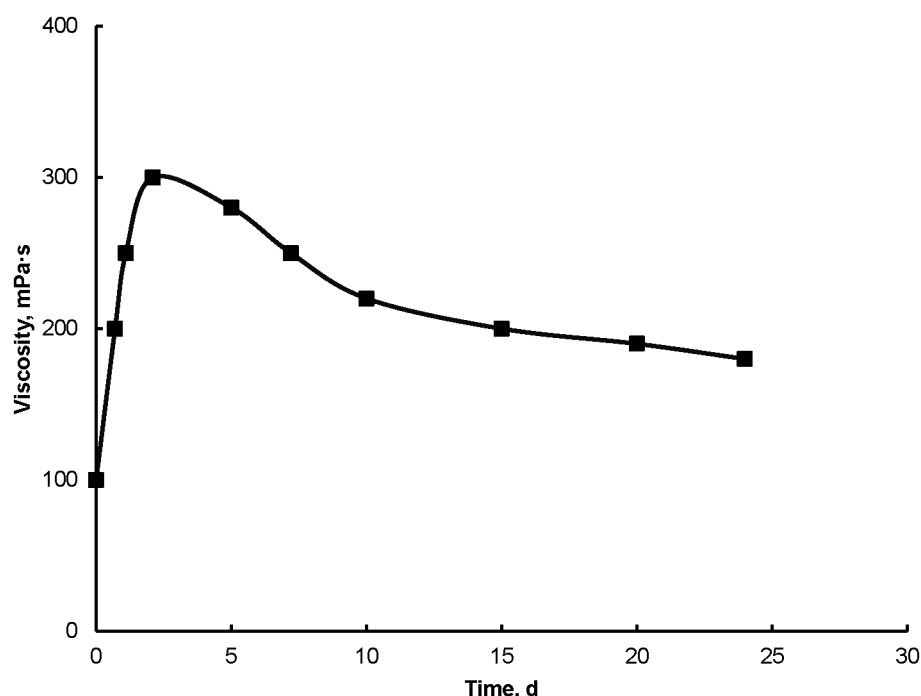


Figure 3—Curve of viscosity change of HVAP aqueous solution over time.

SMG Performance Evaluation Experiments

Following the validation of HVAP, the performance of SMG—the second core component of the composite system—was evaluated to assess its ability for deep profile modification.

Sand-Interbedded Model Sweep Expansion and Displacement Experiment. To study SMG's mesoscopic action mechanism, sand-interbedded model experiments were conducted (Fig. 4). The experimental results show that SMG particles can selectively enter high-permeability layers, where they migrate, swell, and retain, forming a dynamic temporary plugging effect. This prompts subsequent displacement fluids to divert into medium and low-permeability layers, thereby displacing the remaining oil therein.

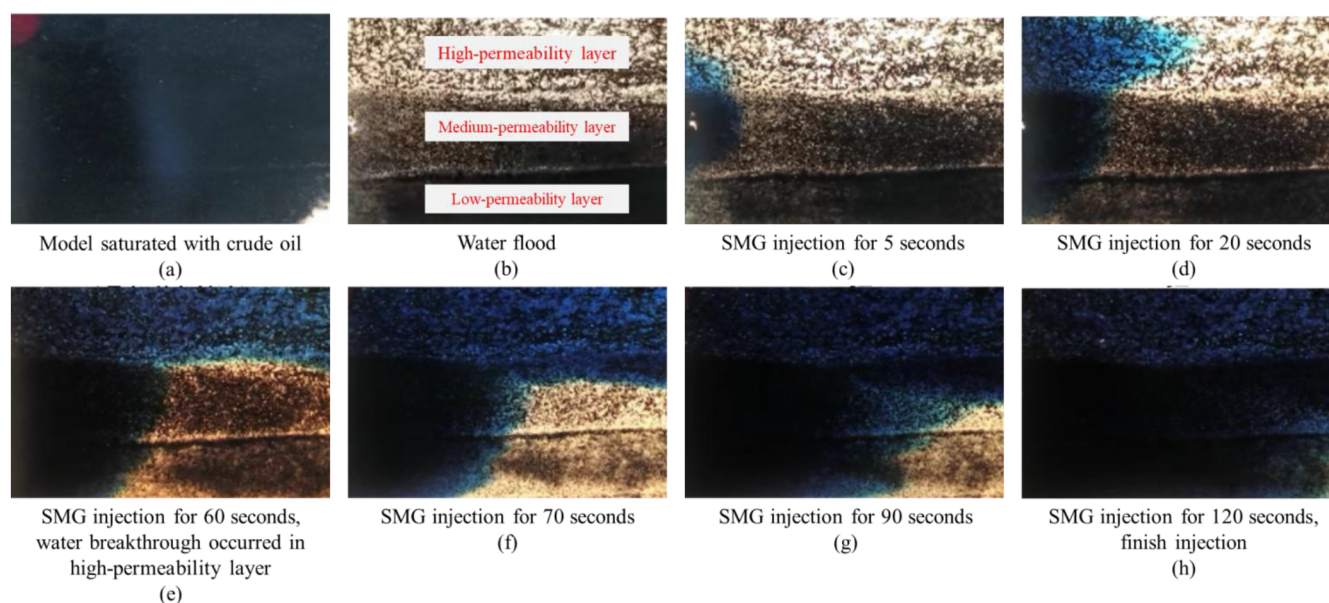


Figure 4—Experimental Study on SMG's Mesoscopic Mechanism.

Pore-Throat Matching Experiment. To investigate the matching relationship between SMG and pore throats, the experiment employed microporous filter membranes with different pore sizes, and the study was conducted by testing the concentration changes of SMG with different particle sizes before and after filtration. The results show that nano-sized SMG exhibits excellent plugging effect on pore throats within 1.0 μm in diameter, while micron-sized SMG has a better plugging effect on pore throats within 5 μm in diameter. Considering the characteristic that the original pore throat size of Y19 Oilfield is 0.83 μm , the selection of nano-sized SMG is more suitable for this oilfield. Figs. 5 and 6 are scanning electron microscope (SEM) images of the matching status between nano-sized and micron-sized SMG particles and pore throats, respectively.

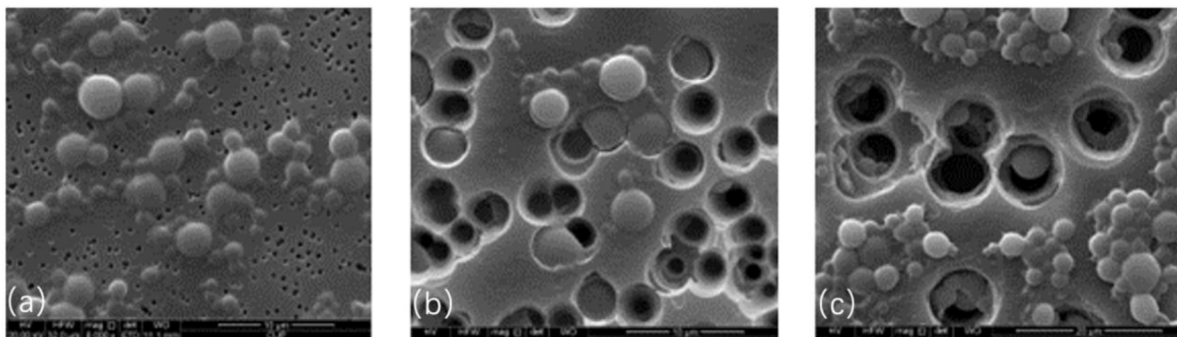


Figure 5—Microscopic Images of Matching between Nano-sized SMG Particles and Pore Throats (a) 1.0 μm nuclear pore membrane; (b) 1.5 μm nuclear pore membrane; (c) 5 μm nuclear pore membrane.

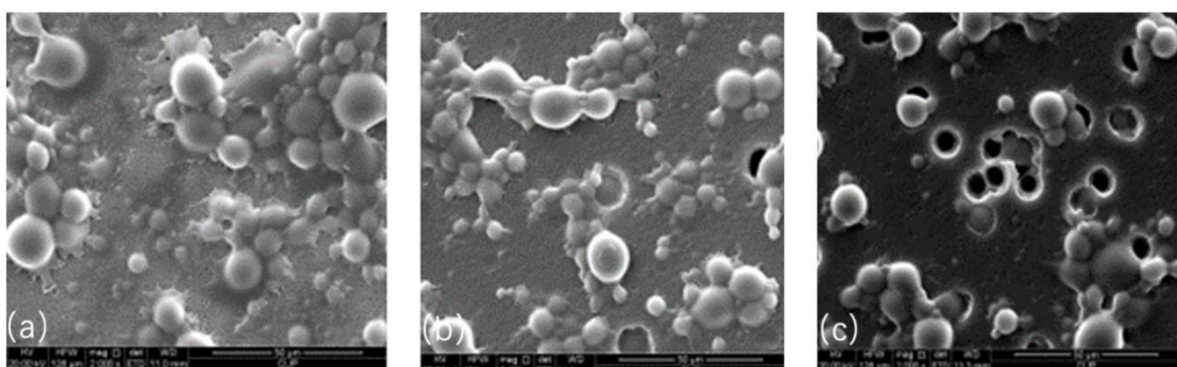


Figure 6—Microscopic Images of Matching between Micron-sized SMG Particles and Pore Throats (a) 1.2 μm nuclear pore membrane; (b) 5 μm nuclear pore membrane; (c) 7 μm nuclear pore membrane.

NanoIE Performance Evaluation Experiments

Interfacial Tension Evaluation. Experimental equipment: TX500 series interfacial tensiometer, as shown in Fig. 7.



Figure 7—TX500C Series Interfacial Tension Meter.

Experimental conditions: Temperature: 45°C; Produced water from Y19 Oilfield; Crude oil from Y19 Oilfield.

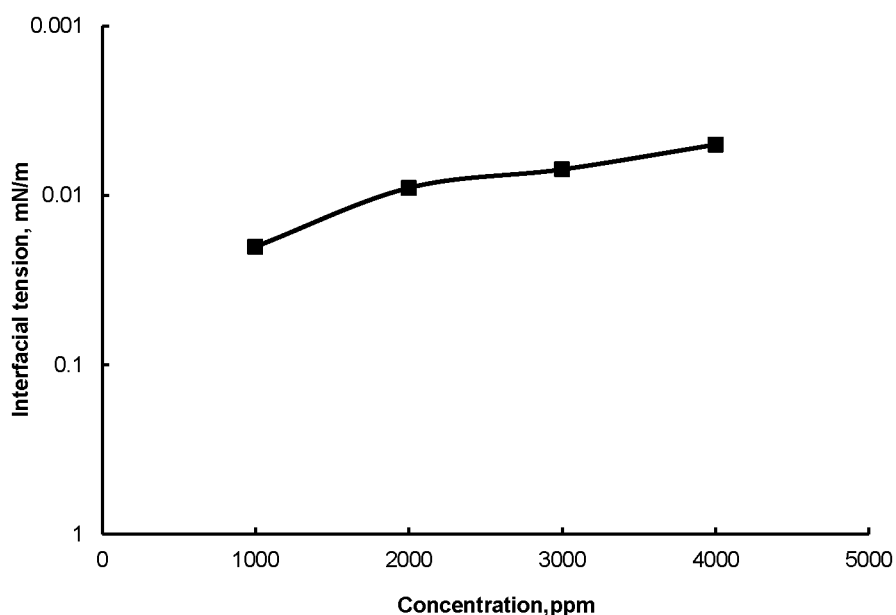


Figure 8—Interfacial tension of NanoIE with different concentrations (prepared with produced water).

Experimental results: When the concentration of NanoIE exceeds 1500 ppm, the interfacial tension of oil-water samples in the test area can reach the order of 10^{-2} ~ 10^{-3} mN/m. From an economic perspective, 1500-3000 ppm is a more reasonable application concentration.

Natural Cores Displacement Experiment

Natural Cores displacement experiments of Y19 Oilfield were conducted for the HVAP + SMG + NanoIE composite system. Through the displacement experiments, the oil displacement efficiency of the composite system on cores with different permeabilities was tested, and the reservoir adaptability of the composite system was evaluated (Table 5).

Table 5—Experimental Scheme for Oil Displacement Efficiency of Natural Cores with Different Permeabilities

No.	Core data				Core flow scheme	Purpose
	Core size	Porosity, %	Permeability, mD	Original oil saturation, %		
High permeability	6.51*2.52cm	17.7	189.3	60.7	Concentration: 0.2% HVAP + 0.3% SMG + 0.3% NanoIE Injection volume: 1 PV Water + 0.5 PV HVAP + 1.0 PV SMG + 1.0 PV NanoIE + 1.0 PV Water Injection rate: 0.03 ml/L	Testing the oil displacement efficiency of the composite system in simulated high- and low-permeability heterogeneous reservoirs
Low permeability	6.50*2.51cm	16.1	23.8	56.0		

From the experimental results, the composite system exhibits good oil displacement effect under the reservoir conditions of Y19 Oilfield. This composite system can form effective plugging on high-permeability layers, significantly inhibit the rise of water cut, expand the swept volume of water flooding, and activate low-permeability layers. Through the organic coupling of the sweep expansion system and the targeted oil displacement system, the oil displacement efficiency of low-permeability reservoirs is improved by 28.64%, and that of high-permeability reservoirs is improved by 21.1%. After applying the HVAP + SMG + NanoIE sweep control & targeted oil displacement composite system, the overall oil displacement efficiency of high- and low-permeability cores is 28.66% higher than that of water flooding, with a sweep coefficient of 60% and an oil recovery improvement of 17.2%.

Table 6—Experimental Results of Oil Displacement in Parallel Dual-Tube Natural Cores from Y19 Oilfield

Scheme		Core permeability, mD	Porosity, %	Oil displacement efficiency/oil displacement efficiency improvement value, %	
				Water flooding	Chemical flooding + secondary water flooding
HVAP + SMG + NanoIE	High permeability	189.3	17.7	30.33	28.66
	Low permeability	23.8	16.1		

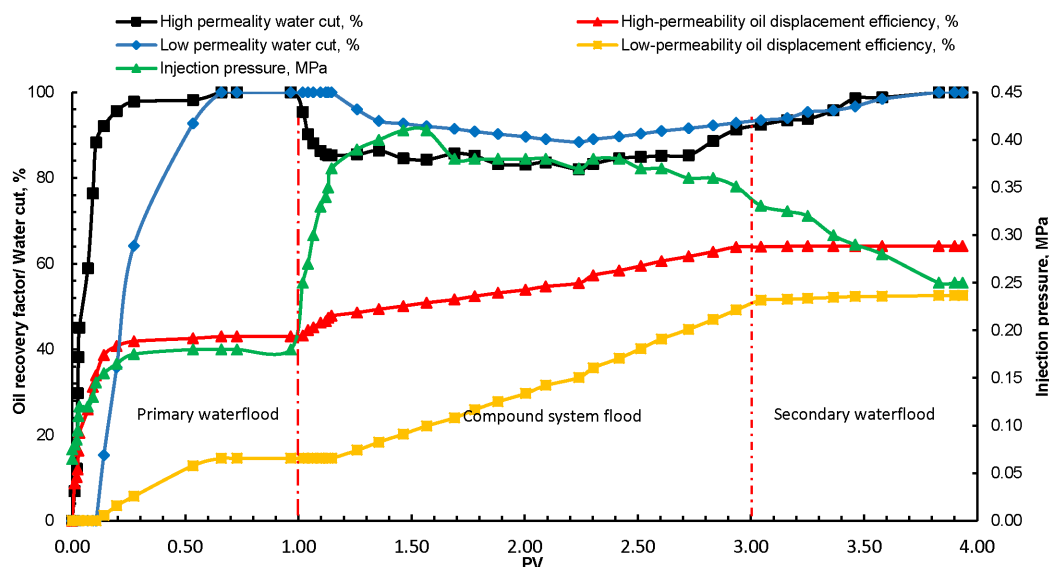


Figure 9—Oil Recovery/Water Cut Curves of High- and Low-Permeability Parallel Experiment with Natural Cores in Composite System.

Scheme Optimization Design

The pilot test covers 4 water injection wells and 18 production wells. At the initial production stage, the daily fluid production was 183.5 m³, daily oil production was 124.5 tons , and water cut was 20.2%. Currently, the daily fluid production is 137.9 m³, daily oil production is 21.7 tons, water cut is 84.4%, and the recovery degree of geological reserves is 22.15%.

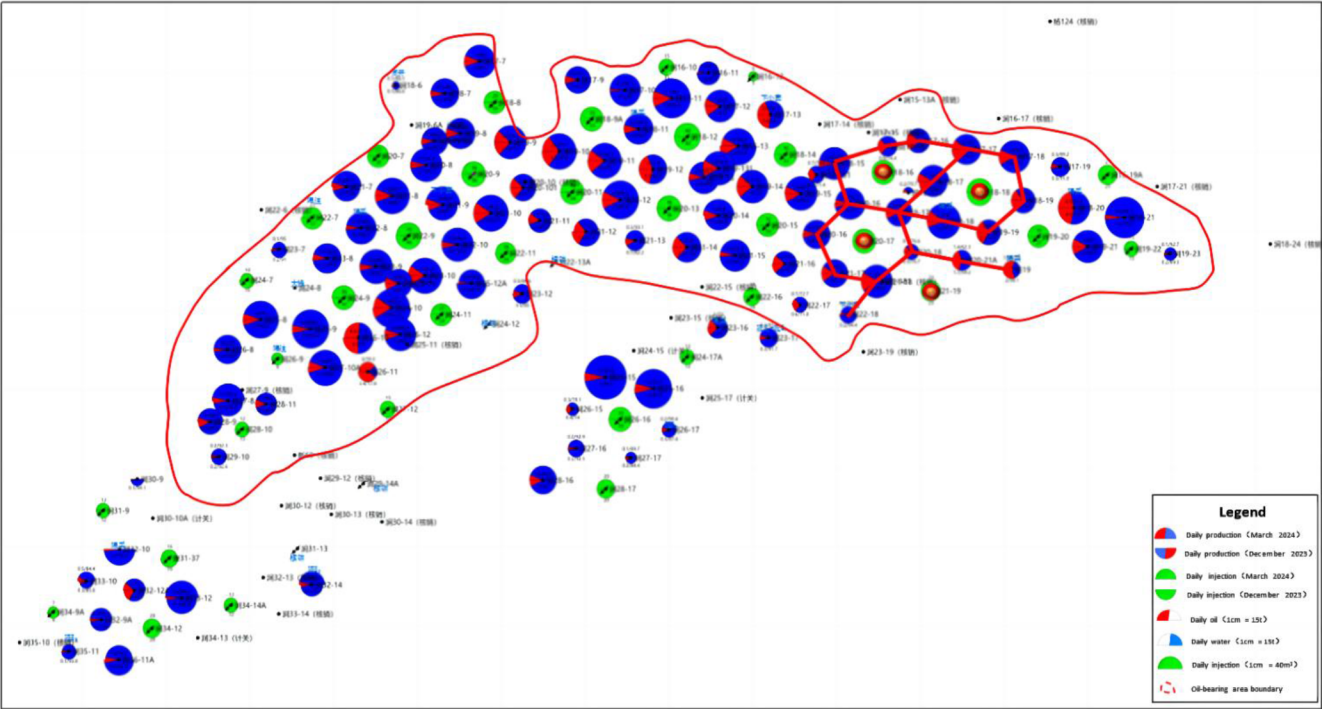


Figure 10—Well Deployment Map of the Pilot Test Area.

Four sweep control and targeted oil displacement schemes were designed, with injection volumes ranging from 0.05 to 0.2 pore volume (PV) (Table 7). All schemes used the optimal concentrations of HVAP (0.2%), SMG (0.3%), and NanoIE (0.3%). Multi-Dimensional Simulation (MDS) chemical flooding software was then used to predict oil recovery, generating a curve of injection volume vs. incremental oil recovery (Fig. 11).

Table 7—Oil Recovery Improvements Corresponding to Different Slug Schemes.

Injection Period: five years				
NO.	Scheme, PV	Slug design	Oil recovery factor , %	Incremental of oil production, t
1	0.05	0.01PV HVAP + 0.02PV SMG + 0.02PV NanoIE	5.2	18,310
2	0.10	0.02PV HVAP + 0.04PV SMG + 0.04PV NanoIE	7.5	24,683
3	0.15	0.03PV HVAP + 0.06PV SMG + 0.06PV NanoIE	10.1	29,625
4	0.20	0.04PV HVAP + 0.08PV SMG + 0.08PV NanoIE	10.7	31,384

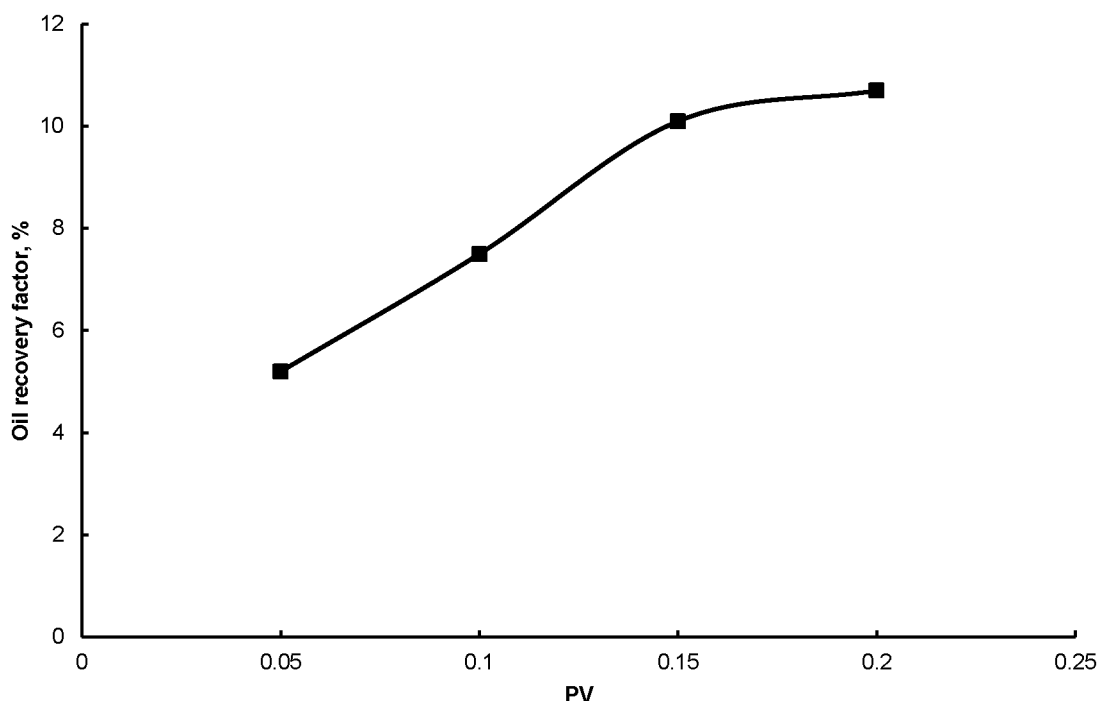


Figure 11—Relationship curve between injected volume PV and oil recovery factor.

Cost-Benefit analysis

In line with the optimized scheme, the chemical flooding will be implemented over 5 years, with a total chemical injection volume of 0.15 pore volume (PV) and an expected incremental oil production of 29,625 tons. To assess the project's economic resilience, a sensitivity analysis of oil price fluctuations was conducted.

The results show that when the oil price is in the range of 40 - 80 USD per barrel, the Return on Investment falls between 1:2.24 and 1:4.48, fully demonstrating that the project possesses favorable economic viability under different market oil price conditions, with stable and considerable revenue expectations.

Table 8—Cost-Benefit analysis and calculation table.

Item	Parameters	Value
Input, K USD	Scheme Design Fee	100
	Chemical Reagent Fee	3,000
	Injection Construction Fee	500
	Testing Fee	70
	Surface Engineering Renovation Fee	30
	Total input, K USD	3,700
Output, K USD	PV injected	0.13
	Average water cut decrease, %	7
	Average daily oil increase, %	15
	Cumulative oil incremental, t	29,625
	40 USD/bbl	8,295
	50 USD/bbl	10,369
	60 USD/bbl	12,443

Item	Parameters	Value
Return on Investment	70 USD/bbl	14,516
	80 USD/bbl	16,590
	40 USD/bbl	1:2.24
	50 USD/bbl	1:2.80
	60 USD/bbl	1:3.36
	70 USD/bbl	1:3.92
	80 USD/bbl	1:4.48

Application Effects

Field testing was conducted from August 2024 to June 2025, and the technology delivered notable results (Figs. 12–14):

- a. The average pressure of the 4 injection wells rose from 3.6 MPa to 5.9 MPa (a 65.4% increase), significantly improving reservoir sweep efficiency;
- b. The test area's daily oil production increased from 31.2 tons (pre-test) to a peak of 42.8 tons (37.4% growth) and currently stabilizes at 36.7 tons—17.8% higher than pre-test levels.
- c. The central well's daily oil production increased from 3.1 tons (pre-test) to a peak of 5.1 tons (63.9% growth) and currently stabilizes at 4.9 tons—54.2% higher than pre-test levels.
- d. In terms of water cut improvement, the comprehensive water cut in the test area decreased from 79.3% to 74.0%, a decrease of 5.3 percentage points; the water cut of the central well decreased from 79.4% to 65.3%, a decrease of 14.1 percentage points.

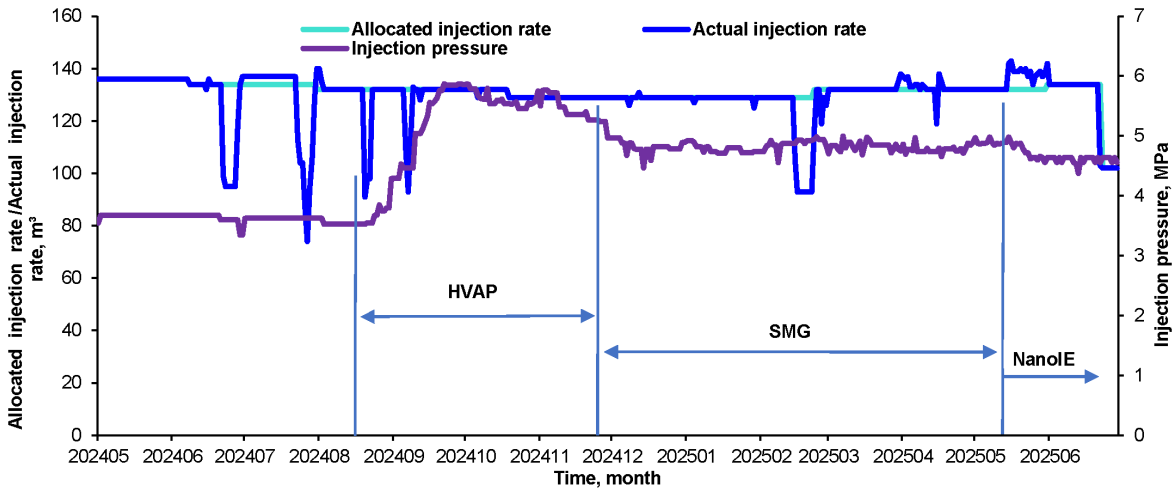


Figure 12—Injection curves of the Pilot Test Area.

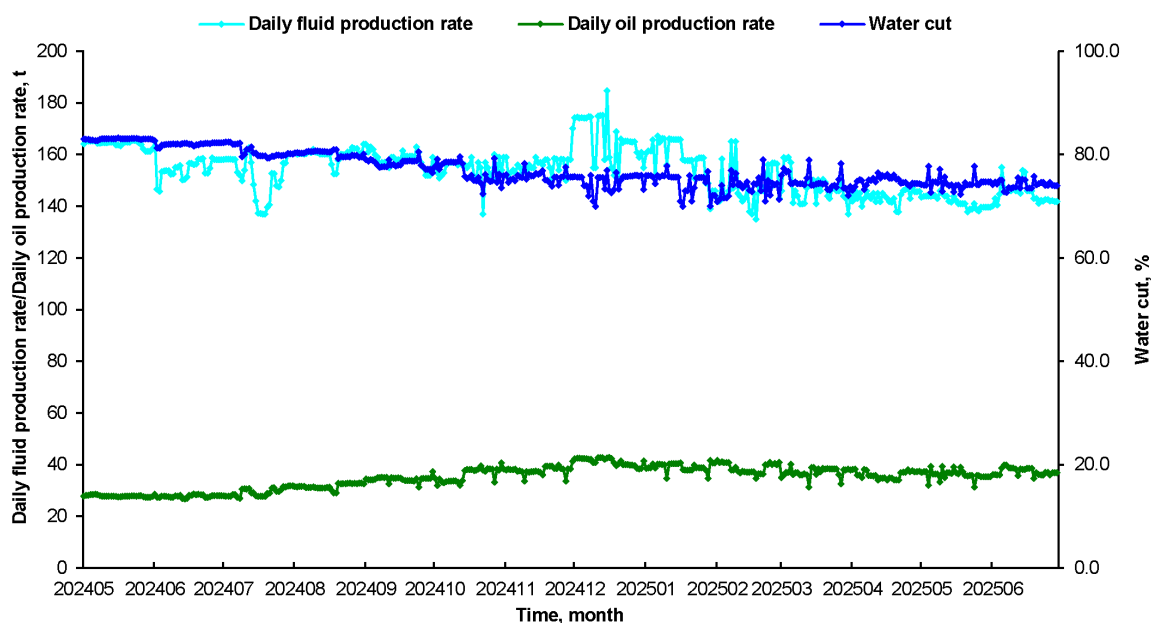


Figure 13—Fluid Production, Oil Production and Water Cut Curves of the Pilot Test Area.

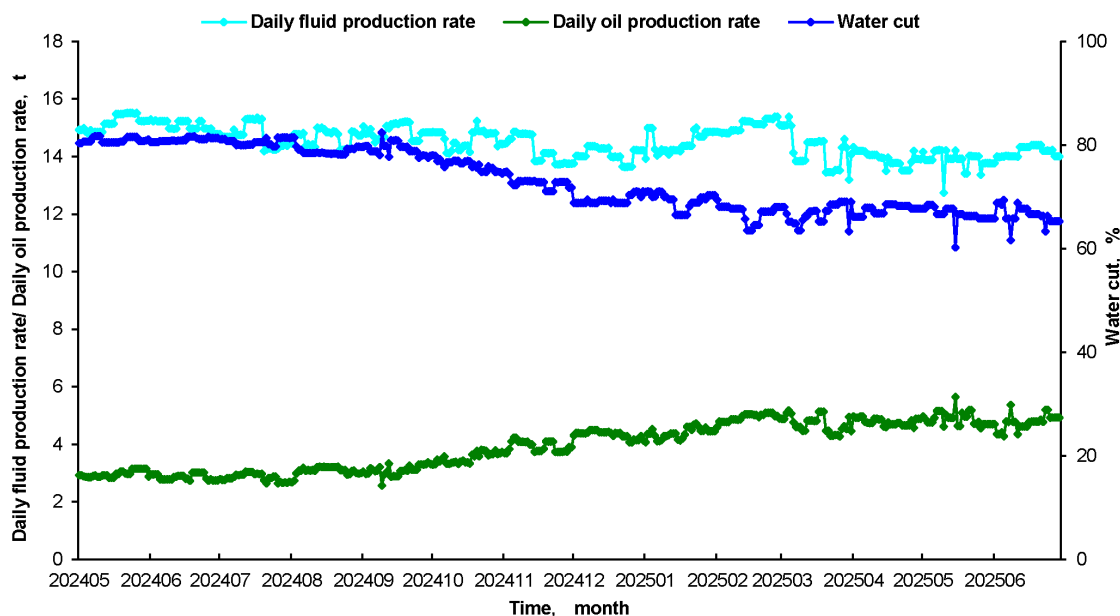


Figure 14—Fluid Production, Oil Production and Water Cut Curves of the central production well.

The synergistic effect of water control and oil increment is significant, and the cumulative oil increment has reached 2488 tons.

Technically, through the synergistic effect of the threefold mechanism of "blocking preferential water flow channels - adjusting seepage profile - improving mobility ratio", the sweep efficiency and displacement efficiency are effectively enhanced. Economic evaluation shows that under the condition of an oil price of \$60 per barrel, the input-output ratio reaches 1:3.6, which has stronger reservoir adaptability (especially suitable for low-permeability sandstone reservoirs) and economic feasibility compared with traditional polymer flooding. This technology provides a solution with both technical innovation and commercial value for the high water cut development stage of low-permeability sandstone reservoirs, and offers practical support for potential tapping, efficiency improvement and sustainable development of mature oilfields.

Conclusions

1. To address the Y19 oilfield's challenges—rapid water cut rise and deteriorating development performance caused by ineffective injected water circulation—this study proposes a sweep control and targeted oil displacement technology based on a HVAP-SMG-NanoIE composite system as the solution. This technology can simultaneously solve the dual bottlenecks of uneven macroscopic displacement and low microscopic displacement efficiency.
2. Laboratory evaluations and core displacement experiments confirm that this technical system is well-compatible with the Y19 reservoir. Moreover, numerical simulation-based scheme optimization shows that the technology achieves optimal displacement efficiency when the total injection volume is 0.15 PV.
3. A field test of the optimized scheme was launched in August 2024, and as of June 2025, it has delivered remarkable results: the test area's daily oil production increased by 17.8%, and the water cut decreased by 5.3 percentage points; the daily oil production of the central well increased by 54.2%, and its water cut decreased by 14.1 percentage points. Compared with traditional polymer flooding, this technology offers stronger reservoir adaptability and economic feasibility, providing an innovative technical path for the high-water-cut development stage of low-permeability sandstone reservoirs.

The sweep control and targeted oil displacement technology has achieved remarkable results in the field application of the Y19 low-permeability reservoir, providing technical support for the efficient development of low-permeability oil reservoirs. However, this technology still has obvious application limitations in complex oil reservoirs where three types of extreme conditions coexist, namely ultra-high temperature ($>130^{\circ}\text{C}$), ultra-high salinity ($>30\times 10^4\text{mg/L}$), and ultra-low permeability ($<1\text{mD}$). These limitations are mainly manifested in issues such as the decreased stability of oil displacement agents, insufficient injection capacity, and weakened targeted effect. In response to the development needs of such special oil reservoirs, future efforts should focus on improving the core performance of products by optimizing molecular structure design and formula systems. Additionally, a combined approach of laboratory simulation experiments and on-site pilot tests should be adopted to systematically verify the adaptability and application potential of this technology, thereby providing technical guarantees for the efficient production of complex oil reservoirs.

References

1. Han Dakuang, Discussions on concepts, countermeasures and technical routes for the redevelopment of high water cut oilfields [J]. *Petroleum Exploration and Development*, 2010, **37**(5):583–591.
2. Hu, Jia, and Ni Xiaolong. "Research Progress on the Adaptability of Reservoir for Polymer flood." *Oil Gas Field Surface Engineering*, vol. **29**, no. 9, 2010, pp. 36–38.
3. Wang, Fang, et al "Formation mechanism and location distribution of blockage during polymer flooding." *Journal of Petroleum Science and Engineering* **194** (2020): 107503.
4. Liu, Li, et al "Experimental study on enhanced oil recovery of the heterogeneous system after polymer flooding." *Processes* **11.10** (2023): 2865.
5. Juárez-Morejón, J. L., et al "A new approach to polymer flooding: Effects of early polymer injection and wettability on final oil recovery." *SPE Journal* **24.01** (2019): 129–139.
6. Lotfollahi, Mohammad, et al "Mechanistic simulation of polymer injectivity in field tests." *SPE Journal* **21.04** (2016): 1178–1191.
7. El-Karsani, Khalid Saad, Ghaithan A. Al-Muntasheri, and Ibelwaleed A. Hussein. "Polymer systems for water shutoff and profile modification: a review over the last decade." *SPE Journal* **19.01** (2014): 135–149.

8. Shen, et al "Study of enhanced-oil-recovery mechanism of alkali/surfactant/polymer flooding in porous media from experiments." *SPE journal* 14.02 (2009): 237–244.
9. Xingcai, Wu, et al "A New IOR Method for Mature Waterflooding Reservoirs: ‘Sweep Control Technology’." SPE-171485-MS, 2014.
10. Xingcai, Wu, Hong biao Wu, Zhong yu Bu, et al, An Innovative EOR Method For Waterflooding Heterogeneous Oilfield ---- Graded Diversion-flooding Technology and Verification by Field Comparison Tests, SPE-189247-MS, 2017
11. Xingcai, Wu. Chunming Xiong, et al A novel particle-type polymer and IOR/EOR property evaluation, SPE177421, 2015
12. Xingcai Wu, Debin Qu, Hanbing Xu, et al, The Economic Analysis and Application Strategies of EOR Technology in Low-Oil-Price Period, SPE-182145-MS, 2016
13. Xingcai Wu, Ying Xiong, Ying Wu, et al, Successful Field Application of a New Polymer Flooding Technique Suitable for Mature Oilfields against Low-oil-Price Background -- A Case Study from East China, SPE-185524-MS, 2017